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ELECTRONIC DEVICE AND METHOD FOR REPRESENTING VISUAL OBJECT IN VIRTUAL ENVIRONMENT

Abstract

An electronic device may include a display, and a processor. The processor may be configured to display a virtual environment comprising a visual object through the display. The processor may be configured to obtain voice information with respect to the visual object while the virtual environment is displayed. The processor may be configured to identify at least one word from the voice information. The processor may be configured to identify that a first word from among the at least one word corresponds to a reference value. The processor may be configured to change the first word to a second word on the basis that the first word corresponds to the reference value. The processor may be configured to display an animation representing that at least one phrase comprising the second word is uttered, in the virtual environment through the visual object and the display.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of International Application No. PCT/KR2023/014320, filed on Sep. 20, 2023, in the Korean Intellectual Property Receiving Office, and claiming priority to Korean Patent Application No. 10-2022-0158230 filed Nov. 23, 2022, and Korean Patent Application No. 10-2022-0173757 filed on Dec. 13, 2022, the disclosures of which are all hereby incorporated by reference herein in their entireties.

BACKGROUND

Technical Field

[0002] Certain example embodiments may relate to an electronic device and a method for representing a visual object in a virtual environment.

Background Art

[0003] In order to provide enhanced user-experience, an electronic device that provides a virtual reality (VR) service displaying a virtual space or an augmented reality (AR) displaying information generated by a computer in association with an external object in the real-world is being developed. The electronic device may include a wearable device. For example, the wearable device may include a VR device, an AR glass, and a head-mounted device (HMD).

SUMMARY

[0004] A wearable electronic device is provided. An example electronic device may include a display. The electronic device may include a processor comprising processing circuitry. The processor, comprising one or more processors, may be configured to display, via the display, a virtual environment including a visual object. The processor may be configured to obtain voice information with respect to the visual object, while the virtual environment is displayed. The processor may be configured to identify at least one word from the voice information. The processor may be configured to identify that a first word from among the at least one word corresponds to a reference value. The processor may be configured to, based on the first word corresponding to the reference value, change the first word to a second word. The processor may be configured to display, via the display, an animation representing uttering the at least one word including the second word through the visual object in the virtual environment.

[0005] A method performed by an electronic device is provided. An example method may include displaying, via a display of the electronic device, a virtual environment including a visual object. The method may include obtaining voice information with respect to the visual object while the virtual environment is displayed. The method may include identifying at least one word from the voice information. The method may include identifying that a first word from among the at least one word corresponds to a reference value. The method may include, based on the first word corresponding to the reference value, changing the first word to a second word. The method may

include displaying, via the display, an animation representing uttering the at least one word including the second word through the visual object in the virtual environment.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0006] FIG. 1 is a block diagram of an electronic device in a network environment according to various example embodiments.

[0007] FIG. 2A is an example block diagram of a method for representing a visual object in a virtual environment according to an example embodiment.

[0008] FIG. 2B illustrates an example of functional components of an electronic device for representing a visual object in a virtual environment according to an example embodiment.

[0009] FIG. 3 is a flowchart illustrating an example of a method for expressing a visual object in a virtual environment according to an example embodiment.

[0010] FIG. 4 is a flowchart illustrating an example of a method for processing input information according to an example embodiment.

[0011] FIG. 5 illustrates an example of a method for predicting input information including a plurality of words according to an embodiment of the present disclosure.

[0012] FIG. 6 illustrates an example of identifying whether a word corresponds to a reference value according to an example embodiment.

[0013] FIG. 7 illustrates an example of identifying an area of a visual object to change an animation in a virtual environment according to an example embodiment.

[0014] FIG. 8 illustrates an example of identifying an animation of a visual object, based on information about a place in a virtual environment according to an example embodiment.

[0015] FIG. 9 is a flowchart illustrating an example of identifying an animation of a visual object, based on a relationship between the visual object and another visual object, according to an example embodiment.

[0016] FIG. 10A illustrates an example of changing an animation of a visual object in a virtual environment according to an example embodiment.

[0017] FIG. 10B illustrates another example of changing an animation of a visual object in a virtual environment according to an example embodiment.

[0018] FIG. 11 illustrates an example of a method for representing a visual object in a virtual environment according to an example embodiment.

DETAILED DESCRIPTION

[0019] The terms used in the disclosure are merely used to better describe a certain embodiment and may not be intended to limit the scope of other embodiments. A singular expression may include a plural expression, unless the context explicitly dictates otherwise. The terms used herein, including technical and scientific terms, may have the same meanings as those commonly understood by those skilled in the art to which the disclosure pertains. Terms defined in a general dictionary among the terms used in the disclosure may be interpreted as having the same or similar meaning as those in the context of the related art, and they are not to be construed in an ideal or overly formal sense, unless explicitly defined in the disclosure. In some cases, even the terms defined in the disclosure may not be interpreted to exclude embodiments of the disclosure.

[0020] In various examples of the disclosure described below, a hardware approach will be described as an example. However, since various embodiments of the disclosure may include a technology that utilizes both the hardware-based approach and the software-based approach, the various embodiments are not intended to exclude the software-based approach.

[0021] As used in the following description, terms referring to a configuration of a device (e.g., processor, camera, display, module, etc.), terms referring to operational states (e.g., step, operation,

procedure), terms referring to signals (e.g., signal, information, etc.), terms referring to data (e.g., parameter, value, etc.) are exemplified for convenience of explanation. Therefore, the present disclosure is not limited to the terms described below, and other terms having equivalent technical meanings may be used.

[0022] Further, throughout the disclosure, an expression, such as e.g., ‘above (more than)’ or ‘below (less than)’ may be used to determine whether a specific condition is satisfied or fulfilled, but it is merely a description for expressing an example and is not intended to exclude the meaning of ‘more than or equal to’ or ‘less than or equal to’. A condition described as ‘more than or equal to’ may be replaced with an expression, such as ‘more than’, a condition described as ‘less than or equal to’ may be replaced with an expression, such as ‘less than’, and a condition described as ‘more than or equal to and below’ may be replaced with ‘more than and less than or equal to’, respectively. Furthermore, hereinafter, ‘A’ to ‘B’ means at least one of the elements from A (inclusive of A) to B (inclusive of B).

[0023] FIG. 1 is a block diagram of an electronic device in a network environment according to an embodiment.

[0024] FIG. 1 is a block diagram of an electronic device in a network environment **100** according to an embodiment. Referring to FIG. 1, an electronic device **101** in a network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, a memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0025] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation.

According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in a volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in a non-volatile memory **134**.

According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0026] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor

121 while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0027] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0028] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0029] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0030] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0031] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0032] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0033] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0034] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0035] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0036] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0037] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0038] The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0039] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0040] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0041] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band)

to address, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0042] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0043] According to an embodiment, the antenna module **197** may be a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0044] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0045] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra-low latency services using, e.g., distributed computing

or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0046] The metaverse is a compound word of the English word ‘meta’, which means “virtual”, “transcendence,” and so on, and “universe,” which means the universe, and refers to a three-dimensional virtual world where social, economic, and cultural activities such as the real world take place. The metaverse is a concept that has evolved one step further than virtual reality (VR, a state-of-the-art technology that allows people to experience the real world in a computer-generated virtual world), and it is characterized by being able to engage in social and cultural activities such as a real life rather than just enjoying games or virtual reality using avatars. Such metaverse service may be provided in at least two forms. The first is to provide the service to a user using a server, and the second is to provide the service through individual contact between users.

[0047] For example, an environment that provides a metaverse service may include a server that provides the metaverse service, a network (e.g., an access point, AP) connecting the server to each user terminal, and a user terminal that accesses the server through a network to use the service by inputting and outputting the metaverse service to the user. Such a case, the server provides a virtual space so that the user terminal may operate in a virtual space. Further, the user terminal may install a software agent or software application for accessing the virtual space provided by the server to express information provided by the server to the user, or transmit information that the user wants to express in the virtual space to the server. The software agent or software application may be provided directly through the server, downloaded from a public server, or embedded when purchasing a terminal.

[0048] As another example, an environment that provides a metaverse service may include a first user terminal that provides the metaverse service, a network (e.g., an access point, AP) connecting each user terminal, and a second user terminal that connects to a server through the network to input and output the metaverse service to a user, thereby allowing the service to be used. In this case, the first user terminal may serve as a server that provides the metaverse service. In other words, a metaverse environment may be configured only by a connection between the devices.

[0049] As described above, the user terminal may be made of various form factors and may include an output device that provides an image and/or sound to a user, and an input device for inputting information to a metaverse service. For example, the user terminal may include a smartphone, an augmented reality (AR) device (e.g., an AR glass), a virtual reality (VR) device, a mixed reality (MR) device, a video through (VST) device, a TV or projector with input/output capability, a computer, and so on. Hereinafter, in the present disclosure, the user terminal may be referred to as an electronic device or a wearable device (e.g., the electronic device **101** of FIG. 1).

[0050] The avatar may refer to a visual object corresponding to a user of an electronic device or a wearable device in a virtual environment of a metaverse service. As interest in the metaverse grows, crimes using avatars in a virtual environment may increase. A user may use the avatar to utter words and actions that cause sexual shame, disgust, or discomfort to other users. For example, the user may damage other users by displaying inappropriate expressions through the avatar using voice chat, avatar expression, or chat. The inappropriate expression may include swear words, expletives, sexual shame, acts that cause disgust, or makes them feel unpleasant. Hereinafter, embodiments of the disclosure propose a method of preventing the user from being offended by identifying an inappropriate expression in information input in a virtual environment and converting an expression of an avatar that may be generated based on the identified expression into another expression. The avatar may be displayed as a visual object to the user via a display of an electronic device or a wearable device (e.g., electronic device **101** of FIG. 1).

[0051] FIG. 2A is an example block diagram of a method for representing a visual object in a virtual environment according to an example embodiment.

[0052] The virtual environment may represent a virtual space of a metaverse service provided through an electronic device (e.g., the electronic device **101** of FIG. 1). The visual object may represent an avatar in the virtual environment corresponding to the user. The user may include a user of an external electronic device different from the electronic device. Hereinafter, the method of FIG. 2A may be performed with an electronic device (e.g., the electronic device **101** of FIG. 1). For example, the method of FIG. 2A may be performed by the processor of the electronic device (e.g., the processor **120** of FIG. 1).

[0053] FIG. 2A illustrates a simplified block diagram of a method of representing a visual object in a virtual environment. For example, the method may include an input operation **200**, a processing operation **210**, a detection and conversion operation **220**, a generation operation **230**, and an output operation **240**.

[0054] According to an embodiment, in the input operation **200**, the processor may identify information from a user of the electronic device and an external electronic device, or a server related to the virtual environment. The information may be referred to as input information.

[0055] According to an embodiment, the input information may be obtained with respect to a visual object in relation to the virtual environment. The input information may be obtained with respect to a visual object corresponding to a user of the external electronic device, while the virtual environment is displayed.

[0056] According to an embodiment, the input information obtained or received in the input operation **200** may include various types of information. For example, the input information may include text information **201**, video information **202**, and voice information **203**. For example, the text information **201** may represent information in the form of text that may be displayed through a user or a server. Further, the video information **202** may represent information in the form of an image including information on movement of a lip of a user or a visual object corresponding to the user. Further, the voice information **203** may represent information on a voice of a user or a visual object corresponding to the user.

[0057] According to an embodiment, the input information may include information inputted to the electronic device and an external electronic device, or information inputted from a server in which the electronic device and the external electronic device are connected. For example, the text information **201** may include information inputted by a user of the external electronic device through an input device of the external electronic device. The text information **201** may include information inputted from the server to the electronic device. Further, the video information **202** may include information obtained by photographing the user's face of the external electronic device through the camera of the external electronic device (e.g., the camera module **180** of FIG. 1), or information on a face part of the visual object in the virtual environment corresponding to the user of the external electronic device. Further, the video information **202** may include information inputted from the server to the electronic device. For example, the voice information **203** may include information inputted by a user of the external electronic device through an input device of the external electronic device, or information on a voice of a visual object in the virtual environment corresponding to the user of the external electronic device. The voice information **203** may include information input from the server to the electronic device.

[0058] According to an embodiment, in the processing operation **210**, the processor may process the input information. For example, the processing operation **210** may include processing for identifying in a unit of word segment (or word) according to the input information. The processing operation **210** may include text information processing **211**, video information processing **212**, and audio information processing **213**. The unit of the word segment is only defined for convenience of explanation, and embodiments of the present disclosure are not limited thereto. The word segment may indicate a minimum unit capable of identifying an inappropriate expression. For example, the

word segment may include a word including a substantially inappropriate expression. For example, the word segment may indicate a part excluding a postpositional particle in a word segment (or word).

[0059] Referring to the text information processing **211**, when the text information **201** is input, the processor may identify at least one word by normalizing for identifying a character part other than a symbol in the text information **201**. For example, when the text information **201** includes “Hello, nice to meet you{circumflex over ()} {circumflex over ()}”, the processor may identify “Hello” and “nice to meet you” which are character parts, excluding the symbols “,” and “{circumflex over ()} {circumflex over ()}”. The character part may represent natural language. The processor may identify at least one word by separating the natural language for each word, and may store it through a memory, a buffer, or a server for each word.

[0060] Referring to the video information processing **212**, in the case of inputting the video information **202**, the processor may identify at least one word, based on the movement information of the lip included in the video information **202**. For example, the processor may identify the natural language from movement information of a lip of a user or a visual object of the video information **202**, through lip reading. The processor may identify at least one word by separating the natural language for each word, and may store it through a memory, a buffer, or a server for each word. “Based on” as used herein covers based at least on.

[0061] Referring to the voice information processing **213**, when voice information **203** is input, the processor may identify at least one word by obtaining at least one phoneme of the voice information **203**. For example, the processor may identify at least one phoneme from the voice information **203** and may identify the natural language through the at least one phoneme. The processor may identify at least one word by separating the natural language for each word. Alternatively, the processor may identify at least one phoneme from the voice information **203** and may directly identify at least one word from the identified at least one phoneme. The processor may store each word through a memory, a buffer, or a server.

[0062] According to an embodiment, in the detection and conversion operation **220**, the processor may perform a detection operation **221** for identifying an inappropriate part of the word and a conversion operation **225** for changing a part of the word. For example, the processor may perform the detection and conversion operation **220** based on the at least one word identified in the processing operation **210**. Further, FIG. 2A illustrates a process of performing the detection and conversion operation **220** based on at least one word by the processor of the electronic device (e.g., a reception device), but embodiments of the present disclosure are not limited thereto. The processor may receive information on a result of detecting an inappropriate part for at least one word by the processor or server of the external electronic device (e.g., a transmitting device), and may perform an operation of converting an expression based on the information. Specific content related thereto will be described in FIG. 6 below.

[0063] According to an embodiment, in the detection operation **221**, the processor may identify an inappropriate portion of the at least one word identified in the processing operation **210**. For example, the processor may obtain at least one word “Fuxx” **222** by processing the text information **201**. For example, the processor may identify “PEK” **223** through a movement of the lip by processing the video information **202**, and identify at least one word “Fuxx” based on it. The processor may identify the phoneme “FEOK” **224** by processing the voice information **203** and identify at least one word “Fuxx” based on it. In the above example, the processor identifies one word “Fuxx”, but the present disclosure is not limited thereto and may identify at least one word including “Fuxx”. The processor may identify an inappropriate portion included in the at least one identified word. For example, the processor may identify an inappropriate portion by identifying whether a value of at least one word corresponds to a reference value. For example, by identifying that “Fuxx” corresponds to a reference value, the processor may identify that a part of the at least one word is an inappropriate portion. The reference value may include values corresponding to an

expression (e.g., profanity, swearing words, sexually degrading expressions) that is not allowed in the virtual environment and values corresponding to an expression predetermined by the user.

[0064] According to an embodiment, in the conversion operation **225**, the processor may change a value represented by a part of a word. For example, the processor may identify that the value represented by the identified word “Fuxx” **226** corresponds to the reference value. Accordingly, the processor may change the value of “F” **227** of “Fuxx” **226** to the value of “B” **229** and change “Fuxx” **226** to “Buck” **228**. In such a case, changing the value of “F” **227** to the value of “B” **229** may be changed based on an algorithm or a statistical model (e.g., a decision tree or a deep learning model) preset in the processor.

[0065] According to an embodiment, in the generation operation **230**, the processor may generate an animation of the visual object based on the at least one word changed in the conversion operation **225**. The processor may generate an animation representing that the at least one word is uttered. For example, the processor may generate an animation representing that at least one word including “Buck” **228** is uttered.

[0066] According to an embodiment, the processor may identify a part of an area displayed through the display of the electronic device in order to identify the visual object to which the animation is applied. Accordingly, the processor may apply the animation to the visual object within the identified area, and display the visual object to which the animation is applied, via the display. A specific example related thereto will be described in FIG. 7.

[0067] According to an embodiment, the processor may identify an animation of the visual object based on various parameters. For example, the processor may identify a change level of the animation of the visual object based on a place parameter. In the case of a specific place, the processor may change an animation to be applied to the visual object to a high level and express the same, but in the case of a place different from the specific place, an animation to be applied to the visual object may be changed to a low level or expressed without any change. A specific example related thereto will be described with reference to FIG. 8.

[0068] Further, the processor may identify whether the animation is applied and the level of change, based on at least one of a parameter representing a role between visual objects in the virtual environment, a parameter representing an age of the user of the external electronic device, a parameter representing a relationship between the user of the electronic device and the user of the external electronic device, or a parameter for a reputation of the visual object in the virtual environment. Details related thereto will be described with reference to FIGS. 9, 10A, and 10B below.

[0069] According to an embodiment, referring to the output operation **240**, the processor may output a visual object based on the animation obtained in the generation operation **230**. For example, the processor may display an animation of the visual object through a display. The processor may output a visual object **241** based on an animation expressing uttering at least one word including the “Buck” **228** generated in the generation operation **230**. The animation **242** of a mouth part of the visual object **241** may express uttering at least one word including the “Buck” **228**.

[0070] According to an embodiment, the processor may output a voice at the same time of outputting the visual object **241**. For example, the processor may output a voice for a word (e.g., at least one word including “Fuxx” **226**) before the visual object **241** is changed in the conversion operation **225**, or a voice for a word after the change (e.g., at least one word including “Buck” **228**), together with the visual object **241**. In other words, the processor may output the voice with or without changing the voice. However, embodiments of the present disclosure are not limited thereto, and the processor may output only the visual object **241**.

[0071] Even when the processor outputs a voice for the word before changing, in the case of outputting together the visual object **241** to which the animation **242** is applied to the word after the change, the user may identify as uttering of a word different from the word before the changing and

the word after the changing. This is based on the McGurk effect, and its details will be described in detail with reference to FIG. 11. In contrast, when the processor identifies that the user of the electronic device is not looking at the virtual environment, the processor may output a voice for the word after the change. For example, the processor may perform eye-tracking on the user of the electronic device through a sensor, and thus may identify that the user of the electronic device is not looking at the virtual environment. The processor may output a voice for the word after the change in order to prevent an inappropriate expression from being output.

[0072] A general metaverse service may not restrict a user from using an avatar (or a visual object). Accordingly, the user may display inappropriate expressions through the avatar in the virtual environment, such as sexually humiliating or degrading behavior or language that makes other users feel uncomfortable. Other users may not use the metaverse service due to the user's inappropriate expression. An device and method according to an embodiment of the present disclosure may change an animation of the visual object or change an output voice in representing the visual object in the virtual environment. Accordingly, the device and method according to an embodiment of the present disclosure may reduce discomfort factors in the metaverse service provided through the virtual environment and provide an optimized user experience. Furthermore, the device and method according to an embodiment of the present disclosure may increase the user's immersion in the metaverse service provided through the virtual environment.

[0073] FIG. 2B illustrates an example of functional components of an electronic device for representing a visual object in a virtual environment, according to an example embodiment.

[0074] The virtual environment may represent a virtual space of a metaverse service provided through an electronic device (e.g., the electronic device **101** of FIG. 1). The visual object may represent an avatar in the virtual environment corresponding to the user. The user may include a user of an external electronic device different from the electronic device. Hereinafter, the electronic device **101** of FIG. 2B may be understood substantially the same as the electronic device **101** of FIG. 1. For example, components included in the electronic device **101** of FIG. 2B may be controlled by the processor (e.g., the processor **120** of FIG. 1).

[0075] FIG. 2B illustrates examples of functional components for representing a visual object in a virtual environment included in the electronic device **101**. According to an embodiment, the electronic device **101** may include an information acquisition module **250**, a target detection module **255**, an expression conversion module **260**, and an expression generation module **265**. However, the present disclosure is not limited thereto, and the expression conversion module **260** and the expression generation module **265** may be implemented as one functional component. In other words, the functional components of FIG. 2B are only of an example, and the electronic device **101** may include a module capable of performing substantially the same function.

[0076] According to an embodiment, the information acquisition module **250** may obtain information from a user of an external electronic device different from the electronic device **101**, and a server related to the virtual environment in which the electronic device **101** and the external electronic device are connected. The information may be referred to as input information. The input information may be obtained for a visual object corresponding to the user of the external electronic device in relation to the virtual environment.

[0077] According to an embodiment, the information acquisition module **250** may perform processing according to the type of input information acquired. For example, the information acquisition module **250** may perform normalization on text information, identify movement information of a lip for video information, and identify at least one phoneme for voice information. Accordingly, the information acquisition module **250** may identify at least one word segment (word) by performing the above processing. The unit of the word is only defined for convenience of explanation, and embodiments of the present disclosure are not limited thereto. The word segment may represent a minimum unit capable of identifying an inappropriate expression. For example, the word segment may include a word including a substantially inappropriate expression.

For example, the word segment may indicate a part excluding a postpositional particle within a word segment (word). The at least one word may include natural language. For example, the natural language may include Korean, English, and the like. The processing may represent natural language processing.

[0078] According to an embodiment, the information acquisition module **250** may separate the at least one word for each word and store the same in a memory, a buffer, or a server. The at least one word may be used to identify whether an inappropriate expression is included, based on a reference value corresponding to an inappropriate expression.

[0079] According to an embodiment, the information acquisition module **250** may identify information on at least one word processed from an external electronic device (e.g., at least one word and a detection result for an inappropriate expression of the at least one word). For example, the processor may obtain information on the at least one word from an external electronic device (e.g., the external electronic device connected to the virtual environment) through a communication circuit, and identify information on the at least one word through the information acquisition module **250**.

[0080] According to an embodiment, the target detection module **255** may identify an inappropriate expression of the identified at least one word. For example, the target detection module **255** may identify a portion of the at least one word corresponding to a reference value, using the reference value corresponding to the inappropriate expression. The target detection module **255** may identify a portion of the at least one word corresponding to the reference value corresponding to an inappropriate expression “Fuxx” (e.g., a combination of a mechanical word value respectively indicating “F”, “U”, “C”, “K”, and a value indicating an array of the mechanical word values. The at least one word may include one word or a plurality of words (e.g., a sentence). For example, the inappropriate expression may include abusive language, profanity, sexually offensive language, or expressions corresponding to a preset value by the user.

[0081] According to an embodiment, the expression conversion module **260** may change a value for a portion of the at least one word identified based on the reference value. The portion may be one word and may include a character (e.g., a consonant or a vowel) within one word. The expression conversion module **260** may change “F” to “B” for the identified part “Fuxx”. Accordingly, the expression conversion module **260** may identify “Buck”, which is at least one word based on the changed value. The changed value may be a value preset in relation to the virtual environment or set by the user of the electronic device. When the value represented by the portion is a consonant, the changed value may also be a consonant. Alternatively, when the value represented by the portion is a vowel, the changed value may also be a vowel. Accordingly, at least one word based on the changed value may be changed into an expression that does not offend the user of the electronic device.

[0082] According to an embodiment, with respect to a portion of the at least one word identified based on the reference value, the expression conversion module **260** may delete a corresponding value or change the corresponding value to an unrelated value rather than changing it to another related value as described above. For example, the expression conversion module **260** may delete a value represented by a portion of the at least one word identified based on the reference value, in order not to provide mouth shape expression or voice information through the visual object. Alternatively, the expression conversion module **260** may change a value represented by a portion of the at least one word identified based on the reference value to an unrelated value, in order to change the mouth shape expression or voice information through the visual object into an unrelated expression. Changing to the unrelated value may indicate changing the visual object corresponding to the user of the external electronic device into an animal-shaped visual object in the example of FIG. **10B** below.

[0083] According to an embodiment, the expression generation module **265** may generate an animation for a visual object, based on at least one word based on the changed value. For example,

the expression generation module **265** may generate an animation representing that the at least one word based on the changed value is uttered with respect to the visual object. However, the present disclosure is not limited thereto, and an animation representing the behavior of the visual object at the same time of indicating that the at least one word is uttered may be generated. According to an embodiment, the generated animation may be stored in a memory or a server through a biovision hierarchy animation (BVH) file.

[0084] According to an embodiment, the expression generation module **265** may identify a visual object to which the animation is applied, based on various parameters. For example, the parameters may include at least one of a place parameter of the virtual environment, a parameter indicating a role between visual objects in the virtual environment, a parameter indicating the age of the user of the external electronic device, a parameter representing a relationship between the user of the electronic device and the user of the external electronic device, or a parameter for the reputation of a visual object in the virtual environment. Further, although not shown in FIG. 2B, the electronic device **101** may display a visual object based on an animation generated by the expression generation module **265** on a display. For example, the electronic device **101** may display the visual object through the display, based on a 2D animation converted by a 3D renderer through the expression generation module **265**.

[0085] FIG. 3 is a flowchart illustrating an example of a method for representing a visual object in a virtual environment according to an example embodiment.

[0086] The virtual environment may represent a virtual space of a metaverse service provided through the electronic device (e.g., the electronic device **101** of FIG. 1). The visual object may represent an avatar in the virtual environment corresponding to the user. The user may include a user of an external electronic device different from the electronic device. Hereinafter, the method of FIG. 3 may be performed by the electronic device **101** of FIG. 2B. The electronic device **101** of FIG. 2B may be understood substantially the same as the electronic device **101** of FIG. 1. For example, the method of FIG. 3 may be performed by the processor of the electronic device (e.g., the processor **120** of FIG. 1).

[0087] Referring to FIG. 3, in operation **300**, the processor may display a virtual environment including a visual object. For example, the processor may display the virtual environment on a display (e.g., the display module **160** of FIG. 1), using a software application that displays the virtual environment. Although not shown in FIG. 3, the processor may execute the software application prior to the operation **300**. The visual object may refer to an avatar corresponding to the user of the external electronic device. Hereinafter, FIG. 3 illustrates the virtual environment including one visual object as an example, but an embodiment of the present disclosure is not limited thereto. The virtual environment may include at least one visual object. The at least one visual object may correspond to each user of at least one external electronic device.

[0088] In operation **305**, the processor may identify input information. For example, the processor may obtain the input information from a user of the electronic device and an external electronic device, or a server related to the virtual environment. The input information may be obtained with respect to a visual object corresponding to the user of the external electronic device in relation to the virtual environment. Information on the virtual environment may be obtained from the server. The processor may display the virtual environment, based on the information on the virtual environment.

[0089] According to an embodiment, the input information may include various types of information. For example, the input information may include text information, video information, and voice information. For example, the text information may indicate information in the form of text that may be displayed through a user or a server. Further, the video information may indicate information in the form of an image, including information on movement of a lip of a user or a visual object corresponding to the user. Further, the voice information may indicate information on a user or a voice of a visual object corresponding to the user.

[0090] According to an embodiment, the input information may include information inputted to the external electronic device or information inputted from a server which the electronic device and the external electronic device are connected to. For example, the text information may include information inputted by a user of the external electronic device through an input device of the external electronic device. The text information may include information inputted from the server to the electronic device. Further, the video information may include information obtained by photographing a face of the user of the external electronic device through a camera of the external electronic device, or information on a face part of the visual object in the virtual environment corresponding to the user of the external electronic device. Further, the video information may include information inputted from the server to the electronic device. For example, the voice information may include information inputted by a user of the external electronic device through an input device of the external electronic device, or information on a voice of a visual object in the virtual environment corresponding to the user of the external electronic device. The voice information may include information inputted from the server to the electronic device.

[0091] In operation **310**, the processor may identify at least one word based on input information. The unit of the word is only defined for convenience of explanation, and embodiments of the present disclosure are not limited thereto. The word may represent a minimum unit capable of identifying an inappropriate expression. For example, the word may include a word including a substantially inappropriate expression. The word may indicate a portion excluding a postpositional particle in the word. For example, the processor may identify at least one word by processing the input information to extract natural language. The natural language may include Korean, English or the like as a meaning opposite to the machine language. Although not shown in FIG. 3, according to an embodiment, the processor may store at least one identified word in a memory or a buffer.

[0092] In operation **315**, the processor may identify whether a portion of at least one word corresponds to a reference value. For example, the processor may identify whether a portion of the at least one stored word corresponds to a reference value. For example, the portion may include a first word included in the at least one word. The reference value may include a value corresponding to an expression (e.g., an expression that causes swear words, profanity, and sexual shame) that may not be allowed in the virtual environment and a value corresponding to an expression determined in advance by the user. In FIG. 3, a portion of at least one word is described as an example, but the present disclosure is not limited thereto. For example, the whole of at least one word may correspond to the reference value. As described above, the processor may detect such an inappropriate expression by identifying whether a portion of at least one word corresponds to the reference value, based on at least one of the following methods.

[0093] For example, the processor may identify that at least one word corresponds to the reference value, by identifying the user's intention included in the input information. Inappropriate intention of the user may indicate that the input information does not include inappropriate expressions such as direct swearing or profanity, but includes inappropriate language when interpreting the meaning of the input information. For example, when at least one word identified is "out of a normal mental state," the at least one word may be an inappropriate expression for criticizing the other person. Further, when at least one word identified is, for example, "dark, too dark to see," the at least one word may be an inappropriate expression for racial discrimination. Further, when at least one word identified is "Take off your clothes," for example, the at least one word may be an inappropriate expression for sexual harassment. In order to identify an intention to express such an inappropriate expression as described above, the processor may use artificial intelligence to identify that the at least one word is a reference value representing an inappropriate expression. For example, based on the stored data on at least one word, the processor may identify that the intention of the at least one word is a reference value representing an inappropriate expression using deep learning. The deep learning may include text classification.

[0094] Further, using a phonological method, the processor may identify that at least one word

corresponds to a reference value. The phonological method is a method of identifying by converting at least one word into phonological units, in order to identify at least one word that represents substantially the same meaning by adding another letter in an inappropriate expression or changing its consonant or vowel. For example, when at least one word is “feock”, its phonological unit may be defined as /f/, /Λ/, and /k/. This may correspond to /f/, /Λ/, /k/ which is a phonological unit of “fuxx” set as an inappropriate expression. Therefore, the processor may identify that at least one word corresponds to a value indicating “fuxx”, even if it is a value indicating “feock”. Further, for example, when at least one word is “fucc”, its phonological unit may be defined as /f/, /Λ/, /k/. This may correspond to /f/, /Λ/, /k/ which is a phonological unit of “fuxx” set as an inappropriate expression. Therefore, the processor may identify that even if at least one word is a value representing “fucc”, it corresponds to a value representing “fuxx”. Further, for example, when at least one word is “custom-character2custom-character(C2pal)”, phonological units may be defined as “custom-character || custom-character|custom-character”. The processor may identify that the at least one word separated by its phonological units is similar to a reference value (“custom-character|custom-character|custom-character”) of “custom-character(Cipal)”, which is an inappropriate expression.

[0095] Further, the processor may identify whether or not at least one word corresponds to a reference value (e.g., a value predetermined by the user) based on a decision tree. The decision tree may represent an algorithm for identifying whether or not at least one word identified based on a user's predetermined value and input information corresponds. For example, the value determined in advance by the user may correspond to an expression not desired by the user. For example, based on the decision tree, the processor may predict a word to be displayed next to at least one word stored so far, or may identify that the intention is inappropriate. The decision tree may represent an algorithm using N-Gram. The N-Gram may be included in a statistical language model (SLM). Specific details related thereto will be described with reference to FIG. 5 below.

[0096] Further, the processor may differently perform an operation of identifying whether or not at least one word corresponds to the reference value, according to an inappropriate expression corresponding to the reference value. For example, when the reference value is a value determined in advance by the user corresponding to an expression not desired by the user, the processor may identify whether or not it corresponds to the reference value based on the at least one word identified by the electronic device. Since the value determined in advance by the user may vary for each user of the electronic device and may include the user's personal information, the electronic device may directly perform an operation of confirming whether or not it is identified. In contrast, when the reference value is a value corresponding to a swear language or profanity, or an inappropriate intention of a sentence (e.g., a sentence structure containing at least one word), the processor may receive a result including information identifying whether or not at least one word identified by the external electronic device and at least one word identified by the external electronic device correspond to the reference value. The processor may identify whether the reference value corresponds to at least one word, based on the received result. In consideration of the relatively objective nature of the content in a case of swear words, profanity, or inappropriate intent of the sentence, and in consideration of availability of the latency for transmitting data, the external electronic device (e.g., a transmitting device) may perform an operation to confirm whether the identification is performed and transmit a result of the performed operation to the electronic device (e.g., a receiving device). The electronic device may identify that information input to the external electronic device is an inappropriate expression, based on the received result. Specific detail related thereto will be described with reference to FIG. 6.

[0097] In operation 315, when the processor identifies that the at least one word corresponds to the reference value, operation 320 may be performed. In contrast, in operation 315, when the processor identifies that the at least one word does not correspond to the reference value, operation 330 may be performed.

[0098] In operation **320**, the processor may change a value represented by a portion of at least one word. When the portion of at least one word corresponds to a reference value, the processor may change a value of the portion corresponding to the reference value to another value. For example, the processor may change a value of a first word of the at least one word to a value representing a second word. For example, when the at least one word is “what the fuxx”, the processor may change “fuxx” which is a word corresponding to the reference value to “buck”. Accordingly, the processor may identify “what the buck”. The second word may represent a word in which a consonant or vowel is changed from the first word.

[0099] In operation **325**, the processor may obtain an animation expressing that at least one changed word is uttered. The processor may obtain an animation representing that “what the buck” identified in the operation **320** is uttered.

[0100] According to an embodiment, the processor may identify a portion of an area displayed through the display of the electronic device in order to identify a visual object to which the animation is applied. Accordingly, the processor may apply the animation to the visual object in the identified area, and display the visual object to which the animation is applied through the display. A specific example related thereto will be described with reference to FIG. 7.

[0101] According to an embodiment, the processor may identify the animation of the visual object based on various parameters. For example, the processor may identify the animation of the visual object based on the place parameter. The processor may identify whether the animation of the visual object is applied and the level of change, according to the place. A specific example related thereto will be described in FIG. 8. Further, the processor may identify whether to apply the animation and a level of change, based on at least one of a parameter representing a role between visual objects in the virtual environment, a parameter representing an age of a user of the visual object (e.g., a user of the external electronic device) or a setting age of the visual object (e.g., a visual object corresponding to a user of the external electronic device), a parameter representing a relationship between a user of the electronic device and a user of the external electronic device, or a parameter for a reputation of a visual object in the virtual environment. Specific detail related to this will be described with reference to FIGS. 9, 10A, and 10B below.

[0102] In operation **330**, the processor may obtain an animation expressing that at least one word is uttered. For example, the processor may identify that the at least one word does not contain an inappropriate expression, as the processor identifies that a portion of the at least one word does not correspond to the reference value. Accordingly, the processor may obtain the animation expressing that the at least one word is uttered.

[0103] In operation **335**, the processor may display an animation through a visual object. The processor may display, through the visual object, an animation representing utterance of at least one word changed in the operation **325** or an animation representing utterance of at least one word not changed in the operation **335**. The visual object may represent an object in the virtual environment corresponding to a user of the external electronic device of the electronic device.

[0104] According to an embodiment, the processor may output an animation of a visual object and a voice at the same time. For example, the processor may output a voice for a word (e.g., “what the Fuxx”) before the visual object is changed in operation **325** or a voice for a word (e.g., “what the buck”) after the change, together with the visual object. In other words, the processor may output the voice with or without changing the voice. However, embodiments of the present disclosure are not limited thereto, and the processor may output only the visual object.

[0105] Even if the processor outputs a voice for a word before the change, when it outputs a visual object to which an animation is applied for the word after the change together, the user of the electronic device may identify it as utterance of a word different from the word before the change and after the change. This is based on the McGurk effect, and its details will be described with reference to FIG. 11. In contrast, when the processor identifies that the user of the electronic device is not looking at the virtual environment, the processor may output the voice for the word after the

change. For example, the processor may perform eye-tracking on the user of the electronic device through a sensor, and thus may identify that the user of the electronic device is not looking at the virtual environment. The processor may output the voice for the word after the change in order to prevent an inappropriate expression from being output.

[0106] Referring to the above description, the device and method according to an embodiment of the present disclosure may change the animation of the visual object or change the output voice in representing the visual object in the virtual environment. Accordingly, the device and method according to an embodiment of the present disclosure can reduce discomfort factors in the metaverse service provided through the virtual environment and provide an optimized user experience. Further, the device and method according to an embodiment of the present disclosure can increase the user's immersion in the metaverse service provided through the virtual environment.

[0107] FIG. 4 is a flowchart illustrating an example of a method for processing input information according to an example embodiment.

[0108] The virtual environment may represent a virtual space of a metaverse service provided through an electronic device (e.g., the electronic device **101** of FIG. 1). The visual object may represent an avatar in the virtual environment corresponding to the user. The user may include a user of an external electronic device different from the electronic device. Hereinafter, the method of FIG. 4 may be performed by the electronic device **101** of FIG. 2B. The electronic device **101** of FIG. 2B may be understood substantially the same as the electronic device **101** of FIG. 1. For example, the method of FIG. 4 may be performed by the processor of the electronic device (e.g., the processor **120** of FIG. 1).

[0109] FIG. 4 illustrates examples of detailed operations of the operation **305** and the operation **310** of FIG. 3. Accordingly, operations other than operation **400** to operation **410** of FIG. 4 may be understood substantially the same as operation of FIG. 3.

[0110] Referring to FIG. 4, in operation **400**, the processor may identify input information. For example, the processor may identify the input information from a user of the electronic device and an external electronic device, or a server related to the virtual environment. The input information may be obtained with respect to a visual object corresponding to the user of the external electronic device in relation to the virtual environment.

[0111] According to an embodiment, the input information may include various types of information. For example, the input information may include text information, video information, and voice information. For example, the text information may indicate information in the form of text that can be displayed through a user or a server. Further, the video information may indicate information in the form of an image, including information on movement of a lip of a user or a visual object corresponding to the user. Further, the voice information may indicate information on a voice of a user or a visual object corresponding to the user.

[0112] According to an embodiment, the input information may include information inputted to an external electronic device different from the electronic device or information inputted from a server in which the electronic device and the external electronic device are connected. For example, the text information may include information inputted by a user of the external electronic device through an input device of the external electronic device. The text information may include information inputted from the server to the electronic device. Further, the video information may include information obtained by photographing the user's face of the external electronic device through the camera of the external electronic device, or information on a face part of the visual object in the virtual environment corresponding to the user of the external electronic device. Further, the video information may include information inputted from the server to the electronic device. For example, the voice information may include information inputted by a user of the external electronic device through an input device of the external electronic device, or information on a voice of the visual object in the virtual environment corresponding to the user of the external

electronic device. The voice information may include information inputted from the server to the electronic device.

[0113] In operation **405**, the processor may identify at least one word by processing the input information. For example, the processor may identify at least one word by processing the input information to extract natural language. The natural language may include Korean or English in a meaning opposite to that of a machine language.

[0114] For example, the processing may include text information processing. When the text information is input, the processor may identify at least one word by normalizing for identifying a character part other than a symbol in the text information. For example, when the text information includes “Hello, :)”, the processor may identify ‘Hello’, which is a character part, excluding the symbols ‘,’ , ‘:’, and ‘)’. The character part may represent a natural language. The processor may identify at least one word by separating the natural language for each word.

[0115] Further, the processing may include video information processing. When the video information is input, the processor may identify at least one word based on motion information of a lip included in the video information. For example, the processor may identify the natural language through lip reading from the motion information of the lip of a user of video information or a visual object corresponding to the user. The processor may identify at least one word by separating the natural language for each word.

[0116] Further, the processing may include voice information processing. When the voice information is input, the processor may identify at least one word by obtaining at least one phoneme of the voice information. For example, the processor may identify at least one phoneme from the voice information and may identify natural language through the at least one phoneme. The phoneme may represent an abstract minimum unit for distinguishing a meaning of a word in at least one word of voice information uttered by a user or a visual object corresponding to the user. The processor may identify at least one word by separating the natural language for each word.

[0117] In operation **410**, the processor may store at least one word in units of words. For example, the processor may store at least one word identified through the above process through a memory or a buffer for each word. The at least one stored word may be used to identify whether an inappropriate expression is included, based on identifying whether the at least one word corresponds to a reference value.

[0118] FIG. 5 illustrates an example of a method for predicting input information including a plurality of words according to an embodiment of the present disclosure.

[0119] The method of FIG. 5 may be performed by the electronic device **101** of FIG. 2B. The electronic device **101** of FIG. 2B may be understood substantially the same as the electronic device **101** of FIG. 1. For example, the method of FIG. 5 may be performed by the processor of the electronic device (e.g., the processor **120** of FIG. 1).

[0120] FIG. 5 illustrates a specific example of a decision tree among methods of identifying whether at least one word includes an inappropriate expression, in operation **315** of FIG. 3. The decision tree may indicate an algorithm for identifying whether at least one word identified based on input information corresponds to a value determined in advance by a user of the electronic device. For example, a predetermined value by the user may correspond to an expression not desired by the user of the electronic device. Based on the decision tree, the processor may predict a word to be displayed next to at least one word stored so far or identify that its intention is inappropriate. For example, the decision tree may indicate an algorithm using N-Gram. The N-Gram may be included in a statistical language model (SLM).

[0121] Referring to FIG. 5, the processor may identify a first set of words **510** based on input information. The first set of words **510** may represent input information representing a plurality of words other than a single word. The intention of a sentence may be determined based on a relationship between a plurality of words, and thus the first set of words **510** may be identified from the input information representing a plurality of words. For example, the first set of words

510 may include “idiot (intending you)”, “skill”, and “really”. According to an embodiment, the processor may identify whether the predetermined value **520** by the user of the electronic device corresponds to the first set of words **510**. For example, the predetermined value **520** by the user may include “*”, “skill”, and “really”. The character “*” may represent any word. In the above-described example, the processor may identify that the first set of words **510** and the predetermined value **520** substantially correspond to each other. Accordingly, the processor may identify that the first set of words **510** includes an inappropriate expression, or that the word **525** following the first set of words **510** might be an inappropriate expression. Accordingly, the processor may set the first set of words **510** as a target for changing the expression.

[0122] Referring to FIG. 5, the processor may identify a second set of words **530** based on input information. The second set of words **530** may represent input information representing a plurality of words instead of a single word. The intention of a sentence may be determined based on a relationship between a plurality of words, and thus the second set of words **530** may be identified from input information representing a plurality of words. For example, the second set of words **530** may include “idiot (intending you)” and “figure”. According to an embodiment, the processor may identify whether the predetermined value **540** by the user corresponds to the second set of words **530**. For example, the predetermined value **540** by the user of the electronic device may include “idiot/idiots (intending you/your)” and “figure”. In the above-described example, the processor may identify that the second set of words **530** substantially corresponds to the predetermined value **540**. Accordingly, the processor may identify that the second set of words **530** includes an inappropriate expression, or that the word following the second set of words **530** might be an inappropriate expression. Accordingly, the processor may set the second set of words **530** as a target for changing the expression.

[0123] FIG. 6 illustrates an example of identifying whether a word corresponds to a reference value according to an example embodiment.

[0124] FIG. 6 illustrates an example of a method of identifying an inappropriate expression based on information identified by an external electronic device among methods of identifying whether at least one word includes an inappropriate expression, in operation **315** of FIG. 3. The external electronic device may be referred to as a transmitting device, and the electronic device may be referred to as a receiving device. However, an embodiment of the present disclosure is not limited to including only the method of the electronic device, which is a receiving device. For example, the method of the external electronic device, which is a transmitting device, may be included in the embodiment of the present disclosure. For example, the transmitting device or the receiving device may be understood substantially the same as the electronic device **101** of FIG. 2B. The electronic device **101** of FIG. 2B may be understood substantially the same as the electronic device **101** of FIG. 1. For example, the method of FIG. 6 may be performed by the processor of the electronic device (e.g., the processor **120** of FIG. 1).

[0125] Referring to FIG. 6, the processor may identify at least one word. For example, the processor may identify a first word **610**, a second word **620**, and a third word **630** based on the input information. The processor may sequentially identify the first word **610**, the second word **620** and the third word **630** in the order of input time.

[0126] According to an embodiment, after input of the input information related to the first word **610** is completed, the input of the input information related to the second word **620** is started and the same time, the processor may identify whether an inappropriate expression of the first word **610** is included in a first section **615**. For example, the processor may identify the first word **610** based on the input information related to the first word **610** in a first partial section **613** of the first section **615**. The processor may identify whether at least a portion of the first word **610** in a second partial section **616** of the first section **615** corresponds to a reference value representing an inappropriate expression.

[0127] Further, after the input of the input information related to the second word **620** is completed,

the input of the input information related to a third word **630** is started and the same time, the processor may identify whether an inappropriate expression of the second word **620** is included in a second section **625**. For example, the processor may identify the second word **620** based on the input information related to the second word **620** in a first partial section **623** of the second section **625**. The processor may identify whether at least a portion of the second word **620** in a second partial section **626** of the second section **625** corresponds to a reference value representing an inappropriate expression.

[0128] Further, after the input of the input information related to the third word **630** is completed, the input of the input information related to other word after the third word **630** is started and the same time, the processor may identify whether or not an inappropriate expression of the third word **630** is included in a third section **635**. For example, the processor may identify the third word **630** based on the input information related to the third word **630** in the first partial section **633** of the third section **635**. The processor may identify whether at least a portion of the third word **630** in the second partial section **636** of the third section **635** corresponds to a reference value representing an inappropriate expression.

[0129] According to an embodiment, the processor may transmit information on each word and information on the result of identification to a receiving device. For example, after the second partial section **616**, the processor may transmit a result identified in the second partial section **616** (e.g., the first word **610** and information indicating that the first word **610** includes an inappropriate expression) to the receiving device. Further, after the second partial section **626**, the processor may transmit a result identified in the second partial section **626** (e.g., the second word **620** and information indicating that the second word **620** includes an inappropriate expression) to the receiving device. Further, after the second partial section **636**, the processor may transmit a result identified in the second partial section **636** (e.g., the third word **630** and information indicating that the third word **630** includes an inappropriate expression) to the receiving device.

[0130] Referring to the above description, the processor may identify whether the word includes an inappropriate expression, using a latency of recognizing input information related to the word and a latency of transmitting the input information from the transmitting device to the receiving device. Since displaying of the animation for the word is performed by the receiving device, the reference value may include an inappropriate expression corresponding to a relatively objective value. For example, the reference value of FIG. **6** may be a value corresponding to an inappropriate expression according to an abusive language, an expletive, or an intention of a sentence. The latencies may be times available only to the transmitting device which is a device to which information is input. Accordingly, the processor may identify the word based on information related to the word, using the latencies, and identify whether the word includes an inappropriate expression. The processor may transmit the identified result to the receiving device. The receiving device may identify a word (e.g., a first word **610** to a third word **630**) according to the input information input to the transmitting device, and whether the word includes an inappropriate expression, based on the identified result.

[0131] Referring to the above description, the device and method according to an embodiment of the present disclosure may efficiently identify inappropriate expressions using a latency for information transmitted from the transmitting device to the receiving device. In expressing a visual object in a virtual environment, the device and method according to an embodiment of the present disclosure may change an animation of the visual object or change the output voice, based on the result of the identification. Accordingly, the device and method according to an embodiment of the present disclosure may reduce discomfort factors in the metaverse service provided through the virtual environment and provide an optimized user experience. Further, the device and method according to an embodiment of the present disclosure may increase the user's immersion in the metaverse service provided through the virtual environment.

[0132] FIG. **7** illustrates an example of identifying an area of a visual object to change animation in

a virtual environment according to an example embodiment.

[0133] The virtual environment may represent a virtual space of a metaverse service provided through an electronic device (e.g., the electronic device **101** of FIG. **1**). The visual object may represent an avatar in the virtual environment corresponding to the user. The user may include a user of an external electronic device different from the electronic device. Hereinafter, the method of FIG. **7** may be performed through an electronic device (e.g., the electronic device **101** of FIG. **1**). For example, the method of FIG. **7** may be performed by the processor of the electronic device (e.g., the processor **120** of FIG. **1**).

[0134] FIG. **7** illustrates an example of a method of identifying a visual object to which an animation obtained based on at least one word including an inappropriate expression may be applied. At least one word including the inappropriate expression may indicate that at least a portion of the at least one word corresponds to a reference value.

[0135] Referring to FIG. **7**, a user **700** may wear the electronic device **101**. For example, the electronic device **101** may be a wearable device in the form of eyeglasses. The electronic device **101** of FIG. **7** may be understood substantially the same as the electronic device **101** of FIG. **1**. FIG. **7** illustrates the electronic device **101** as a wearable device, but embodiments of the present disclosure are not limited thereto. Embodiments of the present disclosure may also be applied to the electronic device **101** other than the wearable device.

[0136] Referring to FIG. **7**, the user **700** may be served a metaverse service through the electronic device **101**. For example, the virtual environment **710** may represent a virtual space of the metaverse service provided through the electronic device **101**. The virtual environment **710** may include visual objects for a plurality of users. For example, the virtual environment **710** may include a first visual object **711**, a second visual object **712**, and a third visual object **713**.

[0137] According to an embodiment, the processor of the electronic device **101** may identify a visual object to which an animation may be applied. The processor may identify an area **720** viewed by the user **700** through the electronic device **101** and identify a visual object within the area **720**. The processor may identify a second visual object **712** in the area **720**. The second visual object **712** may represent a visual object to which an animation obtained based on at least one word including an inappropriate expression may be applied. For example, the processor may identify at least one word, based on input information obtained for the second visual object **712**, and obtain an animation in response to a portion of the identified at least one word corresponding to the reference value. The animation may be applied to the second visual object **712**. The processor may display the animation through the second visual object **712** on a display of the electronic device **101**.

Referring to the above description, the electronic device **101** may identify information on a direction viewed by the user **700** or information on a gaze, and accordingly, may identify the area **720** in the virtual environment **710**. Further, the electronic device **101** may identify a visual object in the area **720** and apply the obtained animation to the identified visual object.

[0138] Referring to the above description, the device and method according to an embodiment of the present disclosure may convert inappropriate expressions that may be caused by another user in the area **720** viewed by the user **700**. Accordingly, the device and method according to an embodiment of the present disclosure may efficiently use resources by applying animation to a portion (e.g., the second visual object **712**) viewed by the user rather than applying animation to the entire virtual environment **710**. The above method may be referred to as foveated rendering.

[0139] FIG. **8** illustrates an example of identifying an animation of a visual object based on information on a place in a virtual environment according to an example embodiment.

[0140] The virtual environment may represent a virtual space of a metaverse service provided through an electronic device (e.g., the electronic device **101** of FIG. **1**). The visual object may represent an avatar in the virtual environment corresponding to the user. The user may include a user of an external electronic device different from the electronic device. Hereinafter, the method of FIG. **8** may be performed through the electronic device (e.g., the electronic device **101** of FIG. **1**).

For example, the method of FIG. 8 may be performed by the processor of the electronic device (e.g., the processor **120** of FIG. 1).

[0141] FIG. 8 illustrates an example of a method of identifying an animation of a visual object based on information on a space. The information on the space may be one of a plurality of parameters for identifying a visual object to which an animation may be applied. The information on the space may be referred to as a place parameter.

[0142] Referring to FIG. 8, a first example **800** may represent an example of a case where a virtual environment is a classroom in which a class is in progress. A second example **810** may represent an example of a case where a virtual environment is a club in which a party is in progress. According to an embodiment, the processor may identify an animation of the visual object based on a place parameter. The place parameter may be determined differently according to the virtual environment. For example, the place parameter may indicate a value for identifying whether an animation is applied and a change level. Whether to apply may indicate whether to apply an animation of the visual object to the visual object. The change level may indicate a degree of filtering or a degree of change for the animation of the visual object.

[0143] Referring to the first example **800**, when the virtual environment is a classroom, it is necessary to remove unnecessary voices or expressions of visual objects corresponding to other users, and thus the place parameter may indicate that the animation is applied and that the degree of change is relatively high. Accordingly, the processor may identify changing the animations of the visual objects in the virtual environment to a first level. In contrast, referring to the second example **810**, when the virtual environment is a club, it may not be necessary to remove unnecessary voices or expressions of visual objects corresponding to other users, and thus the place parameter may indicate that the animation is not applied or that the degree of change is relatively low. Accordingly, the processor may identify to change the animations of the visual objects in the virtual environment to a second level. The first level may indicate that the degree of change is higher than the second level.

[0144] According to an embodiment, the processor may identify a place parameter through a software application displaying the virtual environment. For example, the processor may identify the place parameter based on data about the place stored in the software application.

[0145] FIG. 9 illustrates an example of identifying an animation of a visual object, based on a relationship between the visual object and another visual object according to an example embodiment.

[0146] The another visual object may represent an avatar in a virtual environment corresponding to the user. The user may include a user of an external electronic device different from an electronic device (e.g., the electronic device **101** of FIG. 1). The visual object may represent an avatar in the virtual environment corresponding to the user of the electronic device. The virtual environment may represent a virtual space of a metaverse service provided through the electronic device. Hereinafter, the method of FIG. 9 may be performed by the electronic device **101** of FIG. 2B. The electronic device **101** of FIG. 2B may be understood substantially the same as the electronic device **101** of FIG. 1. For example, the method of FIG. 9 may be performed by the processor of the electronic device (e.g., the processor **120** of FIG. 1).

[0147] Referring to FIG. 9, in operation **900**, the processor may identify information representing a relationship with another visual object. For example, the processor may obtain information representing a relationship stored in a server related to a software application displaying the virtual environment. The relationship may be an index indicating the intimacy between a user of an electronic device and a user of an external electronic device.

[0148] In operation **905**, the processor may identify whether information representing the relationship is less than another reference value. For example, the other reference value may indicate a threshold level for identifying that the user of the electronic device and the user of the external electronic device are in a close relationship. For example, when the value of information

representing the relationship is greater than or equal to the other reference value, the processor may identify that the user of the electronic device and the user of the external electronic device are in a friendly relationship. The friendly relationship may indicate a case where the user of the external electronic device is a friend or a family member, or already knows the user of the electronic device. Alternatively, when the value of information representing the relationship is less than the other reference value, the processor may identify that the user of the electronic device and the user of the external electronic device are in an unfriendly relationship. In operation **905**, when the processor identifies that the value of information representing the relationship is less than the other reference value, operation **910** may be performed. In operation **905**, when the processor identifies that the value of information representing the relationship is greater than or equal to the other reference value, operation **915** may be performed.

[0149] In operation **910**, the processor may identify an animation of a first change level. The first change level may include a high level of filtering. For example, the processor may identify an animation of the first change level, in order to apply filtering at a high level to a visual object corresponding to the other user who is not intimate with the user of the electronic device.

[0150] In operation **915**, the processor may identify an animation of a second change level. The second change level may include a low level of filtering. For example, the processor may identify an animation of the second change level in order to apply filtering at a low level to a visual object corresponding to the other user who is intimate with the user of the electronic device.

[0151] In operation **920**, the processor may display the identified animation through a visual object. For example, the processor may apply an animation of a level identified in operation **910** or operation **915** to the visual object, and may display the visual object on a display.

[0152] In FIG. **9**, users of the external electronic device are categorized as having a friendly or unfriendly relationship based on another reference value of one, but the embodiment of the present disclosure is not limited thereto. Using a plurality of other reference values, the processor may identify a degree of friendly relationship (or intimacy) with the user of the external electronic device. Further, the processor may vary whether to apply the animation and the change level, based on the information representing the relationship according to the reference value corresponding to an inappropriate expression. For example, in the case of a user of the external electronic device having a low level of intimacy, the processor may identify an animation to limit expressions that cause abusive language, profanity, and sexual shame of the user of the external electronic device. In the case of a user of the external electronic device having an intermediate level of intimacy, the processor may identify an animation to limit expressions such as abusive language, profanity, and the like of the user of the external electronic device or at a level set by the user of the electronic device, and may identify an animation to limit expressions that cause sexual shame. In the case of the user of the external electronic device having a high level of intimacy, the processor may identify an animation of a level set by the user of the electronic device for expressions that may cause swear words, profanity, and sexual shame of the user of the external electronic device. The level set by the user may include whether to apply the animation and the change level.

[0153] Referring to the above description, FIG. **9** illustrates an example of identifying by the processor whether an animation of a visual object is applied and a change level based on information representing a relationship. The processor may identify whether the animation is applied and the change level, based on at least one of a parameter indicating a role between visual objects in the virtual environment, a parameter indicating the age of the user of the external electronic device, or a parameter for the reputation of the visual object in the virtual environment.

[0154] For example, the processor may identify whether or not to apply an animation of each of the visual objects and the change level, based on a parameter representing the role between the visual objects in the virtual environment. For example, with respect to an animation of a visual object that is presenting to visual objects corresponding to the audience, the processor may identify the degree of filtering for expressions that may cause swear words, profanity, and sexual shame of the visual

object presented, at a high level. Alternatively, with respect to an animation of each of the visual objects corresponding to the audience, the processor may identify the degree of filtering for expressions that cause swear words, profanity, and sexual shame of the visual object presented, at a low level. For example, the processor may identify a parameter representing the role through a software application displaying the virtual environment.

[0155] For example, the processor may identify whether to apply the animation of the visual object and the change level, based on a parameter indicating the user's age of the visual object or the setting age of the visual object. This is because when the user of the external electronic device is a minor, other users may be offended by using such an inappropriate expression. For example, when the user of the external electronic device is a minor and the visual object corresponding to the user of the external electronic device is set to be a minor, the processor may identify the degree of filtering at a high level for those expressions such as swear words, profanity, and causing sexual shame with respect to the animation of the visual object. Alternatively, even when the user of the external electronic device is a minor and the visual object corresponding to the user of the external electronic device is set to be an adult, the processor may identify the degree of filtering at a high level for those expressions that cause abusive language, profanity, and sexual shame with respect to the animation of the visual object. For example, even when the user of the external electronic device is an adult and the visual object corresponding to the user of the external electronic device is set to be an adult, the processor may identify the degree of filtering at a high level for those expressions such as abusive language, profanity, and causing sexual shame with respect to the animation of the visual object. For example, when the user of the external electronic device is an adult and the visual object corresponding to the user of the external electronic device is set to be an adult, the processor may identify the degree of filtering for those expressions such as abusive language, profanity, and causing sexual shame with respect to the animation of the visual object, at a low level (or user setting level). The processor may identify the parameter for the user's age of the external electronic device and an age set for the visual object corresponding to the user, through a software application displaying the virtual environment.

[0156] For example, the processor may identify whether to apply the animation of the visual object and the change level, based on the parameter for the reputation of the visual object in the virtual environment. This is because it is necessary to sanction the user when he or she continuously uses such abusive language, profanity, and expressions that cause sexual shame. For example, when there exist multiple records of the user of the external electronic device using such inappropriate expressions through the visual object, the processor may identify the degree of filtering, at a high level, for those expressions that cause abusive language, profanity, and sexual shame with respect to the animation of the visual object. Alternatively, when the user's reputation for the visual object of the external electronic device is good, the processor may identify the degree of filtering for expressions that cause abusive language, profanity, and sexual shame with a low level (or user setting level) with respect to the animation of the visual object. Alternatively, when information on the user's reputation of the external electronic device is not obtained (e.g., when there is no record), the processor may identify the degree of filtering, at a user setting level, for expressions that cause swear words, expletives, and sexual shame with respect to the animation of the visual object. The user setting level may represent a level set by the user of the electronic device. For example, the processor may identify the usage record of the user's visual object of the external electronic device through a software application displaying the virtual environment. Alternatively, the software application may provide information indicating that the user is a user who frequently uses such inappropriate expressions, with respect to the visual object having a poor reputation in the virtual environment. Accordingly, the processor may change the animation of the visual object based on the information. A specific example related to this will be described with reference to FIG. 10B below.

[0157] FIG. 10A illustrates an example of changing an animation of a visual object in a virtual

environment according to an example embodiment. FIG. 10B illustrates another example of changing an animation of a visual object in a virtual environment according to an example embodiment.

[0158] The virtual environment may represent a virtual space of a metaverse service provided through an electronic device (e.g., the electronic device **101** of FIG. 1). The visual object may represent an avatar in the virtual environment corresponding to a user. The user may include a user of an external electronic device different from the electronic device. Hereinafter, the method of FIGS. 10A and 10B may be performed through the electronic device (e.g., the electronic device **101** of FIG. 1). For example, the method of FIGS. 10A and 10B may be performed by the processor of the electronic device (e.g., the processor **120** of FIG. 1).

[0159] FIGS. 10A and 10B illustrate examples of changing an animation of a visual object corresponding to the user of the external electronic device, when the user of the external electronic device utters inappropriately.

[0160] Referring to FIG. 10A, a virtual environment **1000** may be displayed on the display (e.g., the display module **160** of FIG. 1) of the electronic device **101**. The virtual environment **1000** may include a first visual object **1010**, a second visual object **1020**, and a third visual object **1030**. According to an embodiment, while the second visual object **1020** utters, an animation **1025** of the second visual object **1020** may be identified to represent a ridiculous expression. The ridiculous expression may represent a change in a specific facial expression and facial color. In other words, the processor may change the animation **1025** of the second visual object **1020** when the user corresponding to the second visual object **1020** utters an inappropriate expression. Accordingly, the user of the electronic device **101** may feel less discomfort even if the second visual object **1020** utters such an inappropriate expression. Further, the user of the second visual object **1020** may recognize that an inappropriate expression is publicly displayed to other users within the virtual environment **1000** and may not use such an inappropriate expression.

[0161] Referring to FIG. 10B, a virtual environment **1050** may be displayed on the display (e.g., the display module **160** of FIG. 1) of the electronic device **101**. The virtual environment **1050** may include a first visual object **1060**, a second visual object **1070**, and a third visual object **1080**. According to an embodiment, the processor may identify an animation of the user's visual object, based on information on the user's reputation. When the user's reputation of the second visual object **1070** is not so good, an animation of a shape of the second visual object **1070** and an animation of a voice output by the second visual object **1070** may be changed. The animation of the shape may be an animal character (e.g., chicken shape), and the animation of the voice may represent the animal's sound (e.g., cluck). When the reputation is not good, it may indicate that the frequency of use of such inappropriate expressions in the user record of the second visual object **1070** identified through the software application is high. As described above, when the animation of the second visual object **1070** is changed, the user of the second visual object **1070** may have some difficulty in communicating with other users (e.g., the user of the first visual object **1060** and the user of the third visual object **1080**) according to the changed animation, so that the use of the inappropriate expressions may be refrained.

[0162] FIG. 11 illustrates an example of a method for representing a visual object in a virtual environment according to an example embodiment.

[0163] The virtual environment may represent a virtual space of a metaverse service provided through an electronic device (e.g., the electronic device **101** of FIG. 1). The visual object may represent an avatar in the virtual environment corresponding to the user. The user may include a user of an external electronic device different from the electronic device. Hereinafter, the method of FIG. 11 may be performed through the electronic device (e.g., the electronic device **101** of FIG. 1). For example, the method of FIG. 11 may be performed by the processor of the electronic device (e.g., the processor **120** of FIG. 1).

[0164] Referring to FIG. 11, the processor may receive input information **1110**. The input

information **1110** may be obtained with respect to a visual object corresponding to a user of the external electronic device, while the virtual environment is displayed. The processor may identify at least one word **1120** based on the input information **1110**. For example, the at least one word **1120** may include “What the fuxx”. The processor may identify whether the at least one word **1120** includes an inappropriate expression. For example, the processor may identify whether an inappropriate expression is included by comparing the at least one word **1120** with a reference value. The processor may identify that a portion **1130** of the at least one word **1120** corresponds to the reference value. The reference value may indicate a value corresponding to an expression such as abusive language, profanity, or causing sexual shame. The processor may change a value of the portion **1130**. For example, the processor may identify the changed portion **1140** representing “Buck” for the value of the portion **1130** representing “fuxx”. The values of the part **1130** may include a consonant or vowel. When a specific value (e.g., ‘F’) of the existing portion **1130** is a consonant, the changed specific value (e.g., ‘B’) may be a consonant, and when a specific value of the existing portion **1130** is a vowel, the changed specific value may be a vowel. The processor may identify at least one changed word **1150** including the changed portion **1140**. The processor may identify an animation **1165** indicating that at least one word **1150** is uttered, based on the at least one changed word **1150**. The processor may display the animation **1165** through the visual object **1160**. For example, the processor may display the visual object **1160** of the animation **1165** on a display.

[0165] According to an embodiment, the processor may output a voice at the same time of displaying the changed animation **1165** of the visual object **1160**. For example, the processor may identify an animation for a voice for the visual object **1160**, and may output an animation for the identified voice together with the animation **1165** through an output device (e.g., the sound output module **155** of FIG. 1). According to an embodiment, the animation for the voice may represent a voice corresponding to at least one word **1120** identified based on the input information **1110** to be changed. When the animation for the voice is a voice corresponding to the at least one word **1120**, it is combined with the animation **1165** of the visual object **1160**, so that the user of the electronic device may substantially identify the voice as a voice different from the voice corresponding to the at least one word **1120**. For example, when a voice corresponding to the at least one word **1120** outputs “What the fuxx” and the animation **1165** utters “What the buck”, the user of the electronic device may identify it as uttering “What the buck”. The above phenomenon may be referred to as McGurk effect. Accordingly, the user of the electronic device may recognize an inappropriate expression as other sound, so that less discomfort may be felt in using the virtual environment. Alternatively, the animation of the voice may represent a voice corresponding to the changed at least one word **1150**. According to an embodiment, the animation of the voice may be identified according to whether the user of the electronic device looks at the display of the electronic device **101**. When the user of the electronic device looks at the display, even if a word including an inappropriate expression (e.g., at least one word **1120**) is output as a voice as it is, it is combined with an animation **1165** representing that the word (e.g., at least one word **1150**) changed from an inappropriate expression is uttered, so that the user of the electronic device may recognize that other voice is expressed. In contrast, when the user of the electronic device does not look at the display, a word (e.g., at least one word **1120**) including an inappropriate expression is not output as a voice as it is, and the changed word (e.g., at least one word **1150**) is output as a voice so that the user is not offended.

[0166] Referring to the above description, the device and method according to an embodiment of the present disclosure may change an animation of a visual object or change the output voice in expressing the visual object in the virtual environment. Accordingly, the device and method according to an embodiment of the present disclosure can reduce discomfort factors in a metaverse service provided through the virtual environment, thereby providing an optimized user experience. Further, the device and method according to an embodiment of the present disclosure can increase

the user's immersion in the metaverse service provided through the virtual environment.

[0167] As described above, a wearable device **101** may include a display **160**. The wearable device **101** may include a processor **120**. The processor **120** may be configured to display, via the display **160**, a virtual environment including a visual object. The processor **120** may be configured to identify input information obtained for the visual object, while the virtual environment is displayed. The processor **120** may be configured to identify at least one word from the input information. The processor **120** may be configured to identify that a first value represented by a portion of the at least one word corresponds to a reference value. The processor **120** may be configured to change a value represented by the portion of the at least one word from the first value to a second value. The processor **120** may be configured to display an animation representing utterance of the at least one word including the portion indicating the second value, via the display **160** in the virtual environment through the visual object.

[0168] According to an embodiment, the input information may include at least one of first information in which a user of another wearable device different from the wearable device **101** inputs with respect to the virtual object in the virtual environment, or second information displayed in a server related to a software application displaying the virtual environment. The reference value may include a predetermined value by a user of the electronic device or a value representing an expression unavailable in the virtual environment.

[0169] According to an embodiment, the processor **120** may be configured to receive a result of the first information from the another wearable device. The result may include information for representing whether the portion of the at least one word identified from the first information input by a user of the another wearable device corresponds to a value representing an expression unavailable in the virtual environment of the reference value.

[0170] According to an embodiment, the processor **120** may be configured to identify information representing a relationship between the user of the another wearable device corresponding to the another visual object in the virtual environment and the user of the wearable device **101**, using the software application. The processor **120** may be configured to identify a change level of the animation as a first level, in case that the information representing the relationship is less than another reference value. The processor **120** may be configured to identify the change level of the animation as a second level lower than the first level, in case that the information representing the relationship is greater than or equal to the another reference value. The processor **120** may be configured to display the animation of the identified change level through the visual object.

[0171] According to an embodiment, the processor **120** may be configured to identify whether the portion of the at least one word identified from the first information corresponds to a value predetermined by the user of the wearable device **101**.

[0172] According to an embodiment, the processor **120** may be configured to perform a processing of the input information including text information, voice information, and video information. The processing may include identifying the at least one word by normalizing for identifying a character portion other than a symbol in the text information. The processing may include identifying the at least one word by obtaining at least one phoneme for the voice information. The processing may include identifying the at least one word based on movement information of a lip included in the video information.

[0173] According to an embodiment, the value represented by the portion may include a consonant or a vowel. When the first value is a first consonant, the second value may be a second consonant different from the first consonant. When the first value is a first vowel, the second value may be a second vowel different from the first vowel.

[0174] According to an embodiment, the processor **120** may be configured to obtain information on a space of the virtual environment, using a software application displaying the virtual environment. The processor **120** may be configured to identify the animation, based on information on the space of the virtual environment.

[0175] According to an embodiment, the wearable device **101** may further include a speaker. The processor **120** may be configured to output the at least one word including the portion indicating the first value through the speaker, while displaying the animation through the visual object.

[0176] According to an embodiment, the processor **120** may be configured to identify a region being capable of identifying in the virtual environment corresponding to a direction viewed by the user of the wearable device **101**. The animation may be applied to the visual object in the region.

[0177] As described above, a method performed by a wearable device **101** may include displaying a virtual environment including a visual object via a display **160** of the wearable device **101**. The method may include identifying input information obtained for the visual object, while the virtual environment is displayed. The method may include identifying at least one word from the input information. The method may include identifying that a first value represented by a portion of the at least one word corresponds to a reference value. The method may include changing a value represented by the portion of the at least one word from the first value to a second value. The method may include displaying, via the display **160**, an animation representing uttering the at least one word including the second word through the visual object in the virtual environment.

[0178] According to an embodiment, the input information may include at least one of first information in which a user of another wearable device different from the wearable device **101** inputs with respect to another virtual object in the virtual environment, or second information displayed in a server related to a software application displaying the virtual environment. The reference value may include a predetermined value by a user of the wearable device **101** or a value representing an expression unavailable in the virtual environment

[0179] According to an embodiment, the method may include receiving a result of the first information from the another wearable device. The result may include information for representing whether the portion of the at least one word identified from the first information input by the user of the another wearable device corresponds to a value for representing an expression unavailable in the virtual environment of the reference value.

[0180] According to an embodiment, the method may include identifying information representing a relationship between the user of the another wearable device corresponding to the another visual object in the virtual environment and the user of the wearable device **101**, using the software application. The method may include identifying a change level of the animation as a first level, in case that the information representing the relationship is less than another reference value. The method may include identifying the change level of the animation as a second level lower than the first level, in case that the information representing the relationship is greater than or equal to the another reference value. The method may include displaying the animation of the identified change level through the visual object.

[0181] According to an embodiment, the method may include identifying whether the portion of the at least one word identified from the first information corresponds to a value predetermined by the user of the wearable device **101**.

[0182] According to an embodiment, the method may include performing a processing of the input information including text information, voice information, and video information. The processing may include identifying the at least one word by normalizing for identifying a character portion other than a symbol in the text information. The processing may include identifying the at least one word by obtaining at least one phoneme of the voice information. The processing may include identifying the at least one word based on movement information of a lip included in the video information.

[0183] According to an embodiment, the value represented by the portion may include a consonant or a vowel. When the first value is a first consonant, the second value may be a second consonant different from the first consonant. When the first value is a first vowel, the second value may be a second vowel different from the first vowel.

[0184] According to an embodiment, the method may include obtaining information on a space of

the virtual environment using a software application displaying the virtual environment. The method may include identifying the animation based on information on the space of the virtual environment.

[0185] According to an embodiment, the method may include outputting the at least one word including the portion representing the first value via a speaker of the wearable device **101**, while displaying the animation through the visual object.

[0186] According to an embodiment, the method may include identifying a region being capable of identifying in the virtual environment corresponding to a direction viewed by the user of the wearable device **101**. The animation may be applied to the visual object in the region.

[0187] As described above, an electronic device **101** may include a display **160**. The electronic device **101** may include a processor **120**. The processor **120** may be configured to display, via the display **160**, a virtual environment including a visual object. The processor **120** may be configured to obtain voice information with respect to the visual object, while the virtual environment is displayed. The processor **120** may be configured to identify at least one word from the voice information. The processor **120** may be configured to identify that a first word from among the at least one word corresponds to a reference value. The processor **120** may be configured to, based on the first word corresponding to the reference value, change the first word to a second word. The processor **120** may be configured to display, via the display **160**, an animation representing uttering the at least one word including the second word through the visual object in the virtual environment.

[0188] According to an embodiment, the voice information may include at least one of first information in which a user of another electronic device different from the electronic device **101** inputs with respect to the virtual object in the virtual environment, or second information displayed in a server related to a software application displaying the virtual environment. The reference value may include a predetermined value by a user of the electronic device **101** or a value representing an expression unavailable in the virtual environment. The virtual environment may be displayed based on information received from the server. The visual object may represent an avatar corresponding to the user of the another electronic device.

[0189] According to an embodiment, the processor **120** may be configured to receive, from the another electronic device, a result of the first information. The result may include information for representing whether the first word of the at least one word identified from the first information corresponds to the reference value representing an expression unavailable in the virtual environment.

[0190] According to an embodiment, the processor **120** may be configured to identify information representing a relationship between the user of the electronic device **101** and the user of the another electronic device corresponding to the visual object in the virtual environment, using the software application. The processor **120** may be configured to identify a change level of the animation as a first level, in case that the information representing the relationship is less than another reference value. The processor **120** may be configured to identify the change level of the animation as a second level lower than the first level, in case that the information representing the relationship is greater than or equal to the another reference value. The processor **120** may be configured to display the animation of the identified change level through the visual object.

[0191] According to an embodiment, the processor **120** may be configured to identify whether the first word of the at least one word identified from the first information corresponds to a predetermined value by the user of the electronic device **101**.

[0192] According to an embodiment, the processor **120** may be configured to perform a processing of input information including the voice information. The processing may include identifying the at least one word by obtaining at least one phoneme for the voice information. The input information may include text information, the voice information, and video information.

[0193] According to an embodiment, the value represented by a part of the first word may include a

consonant or a vowel. In case that the part of the first word is a first consonant, a portion of the second word corresponding to the part is a second consonant different from the first consonant. In case that the part of the first word is a first vowel, a portion of the second word corresponding to the part is a second vowel different from the first vowel.

[0194] According to an embodiment, the processor **120** may be configured to obtain information on a space of the virtual environment, using a software application displaying the virtual environment. According to an embodiment, the processor **120** may be configured to identify the animation based on the information on the space of the virtual environment.

[0195] According to an embodiment, the electronic device **101** may further comprise a speaker. The processor **120** may be configured to output, via the speaker, the at least one word including the first word, while displaying the animation through the visual object.

[0196] According to an embodiment, the processor **120** may be configured to identify a region being capable of identifying in the virtual environment corresponding to a direction viewed by a user of the electronic device. The animation may be applied to the visual object in the region.

[0197] As described above, a method performed by an electronic device **101** may include displaying, via a display **160** of the electronic device **101**, a virtual environment including a visual object. The method may include obtaining voice information with respect to the visual object, while the virtual environment is displayed. The method may include identifying at least one word from the voice information. The method may include identifying that a first word from among the at least one word corresponds to a reference value. The method may include, based on the first word corresponding to the reference value, changing the first word to a second word. The method may include displaying, via the display, an animation representing uttering the at least one word including the second word through the visual object in the virtual environment.

[0198] According to an embodiment, the input information may include at least one of first information in which a user of another wearable device different from the wearable device **101** inputs with respect to another virtual object in the virtual environment, or second information displayed in a server related to a software application displaying the virtual environment. The reference value may include a predetermined value by a user of the wearable device **101** or a value representing an expression unavailable in the virtual environment. The virtual environment may be displayed based on information received from the server. The visual object represents an avatar corresponding to the user of the another electronic device.

[0199] According to an embodiment, the method may include receiving a result of the first information from the another wearable device. The result may include information for representing whether the first word of the at least one word identified from the first information corresponds to the reference value representing an expression unavailable in the virtual environment.

[0200] According to an embodiment, the method may include identifying information representing a relationship between the user of the electronic device **101** and the user of the another electronic device corresponding to the visual object in the virtual environment, using the software application. The method may include identifying a change level of the animation as a first level, in case that the information representing the relationship is less than another reference value. The method may include identifying the change level of the animation as a second level lower than the first level, in case that the information representing the relationship is greater than or equal to the another reference value. The method may include displaying the animation of the identified change level through the visual object.

[0201] According to an embodiment, the method may include identifying whether the first word of the at least one word identified from the first information corresponds to a predetermined value by the user of the electronic device **101**.

[0202] According to an embodiment, the method may include performing a processing of input information including the voice information. The processing may include identifying the at least one word by obtaining at least one phoneme for the voice information. The input information may

include text information, the voice information, and video information.

[0203] According to an embodiment, the value represented by a part of the first word may include a consonant or a vowel. In case that the part of the first word is a first consonant, a portion of the second word corresponding to the part is a second consonant different from the first consonant. In case that the part of the first word is a first vowel, a portion of the second word corresponding to the part is a second vowel different from the first vowel.

[0204] According to an embodiment, the method may include obtaining information on a space of the virtual environment, using a software application displaying the virtual environment. The method may include identifying the animation based on the information on the space of the virtual environment.

[0205] According to an embodiment, the electronic device **101** may further comprise a speaker. The method may include outputting, via the speaker, the at least one word including the first word, while displaying the animation through the visual object.

[0206] According to an embodiment, the method may include identifying a region being capable of identifying in the virtual environment corresponding to a direction viewed by a user of the electronic device. The animation may be applied to the visual object in the region.

[0207] As described above, a non-transitory computer readable storage medium, when executed by a processor **120** of an electronic device **101** comprising a display **160**, may store one or more programs including instructions that cause to display, via the display **160**, a virtual environment including a visual object. The non-transitory computer readable storage medium, when executed by the processor **120**, may store one or more programs including instructions that cause to obtain voice information with respect to the visual object, while the virtual environment is displayed. The non-transitory computer readable storage medium, when executed by the processor **120**, may store one or more programs including instructions that cause to identify at least one word from the voice information. The non-transitory computer readable storage medium, when executed by the processor **120**, may store one or more programs including instructions that cause to identify that a first word from among the at least one word corresponds to a reference value. The non-transitory computer readable storage medium, when executed by the processor **120**, may store one or more programs including instructions that cause to, based on the first word corresponding to the reference value, change the first word to a second word. The non-transitory computer readable storage medium, when executed by the processor **120**, may store one or more programs including instructions that cause to display, via the display **160**, an animation representing uttering the at least one word including the second word through the visual object in the virtual environment.

[0208] The electronic device according to various embodiments of the disclosure may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. The electronic devices according to an example embodiment are not limited to those described above.

[0209] It should be appreciated that various embodiments and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is

referred to, with or without the term “operatively” or “communicatively”, as “coupled to/with”, or “connected to/with” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0210] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may be interchangeably used with other terms, for example, “logic,” “logic block,” “unit”, “part,” “portion” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC). Thus, each “module” herein may comprise circuitry.

[0211] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., an internal memory **136** or an external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0212] According to an embodiment, a method according to various embodiments may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., a compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0213] According to various embodiments of the disclosure, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

Claims

1. An electronic device comprising: a display; memory comprising one or more storage media storing instruction; and at least one processor comprising processing circuitry, wherein the instructions, when executed by the at least one processor individually or collectively, cause the

electronic device to: display, via the display, a virtual environment including a visual object, obtain voice information with respect to the visual object while the virtual environment is displayed, identify at least one word from the voice information, identify that a first word from among the at least one word corresponds to a reference value, based on the first word corresponding to the reference value, change the first word to a second word, and display, via the display, an animation representing uttering the at least one word including the second word through the visual object in the virtual environment.

2. The electronic device of claim 1, wherein the voice information includes at least one of first information in which a user of another electronic device different from the electronic device has input with respect to the virtual object in the virtual environment, or second information to be displayed in a server related to a software application displaying the virtual environment, wherein the reference value includes a predetermined value by a user of the electronic device and/or a value representing an expression unavailable in the virtual environment, wherein processor is configured so that the virtual environment is displayed based on information received from the server, wherein the visual object represents an avatar corresponding to the user of the another electronic device.

3. The electronic device of claim 2, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: receive, from the another electronic device, a result of the first information, wherein the result includes information for representing whether the first word of the at least one word identified from the first information corresponds to the reference value representing an expression unavailable in the virtual environment.

4. The electronic device of claim 3, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: identify information representing a relationship between the user of the electronic device and the user of the another electronic device corresponding to the visual object in the virtual environment, in case that the information representing the relationship is less than another reference value, identify a change level of the animation as a first level, in case that the information representing the relationship is greater than or equal to the another reference value, identify the change level of the animation as a second level lower than the first level, display the animation with the identified change level through the visual object.

5. The electronic device of claim 2, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: identify whether the first word of the at least one word identified from the first information corresponds to a predetermined value by the user of the electronic device.

6. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: perform a processing of input information including the voice information, wherein the processing includes identifying the at least one word by obtaining at least one phoneme for the voice information, wherein the input information includes text information, the voice information, and video information.

7. The electronic device of claim 1, wherein the value represented by a part of the first word includes a consonant or a vowel, wherein, in case that the part of the first word is a first consonant, portion of the second word corresponding to the part is a second consonant different from the first consonant, wherein, in case that the part of the first word is a first vowel, portion of the second word corresponding to the part is a second vowel different from the first vowel.

8. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: obtain information on a space of the virtual environment by using a software application for displaying the virtual environment, identify the animation based on the information on the space of the virtual environment.

9. The electronic device of claim 1, wherein the electronic device further comprises a speaker, wherein the instructions, when executed by the at least one processor individually or collectively,

cause the electronic device to: output, via the speaker, the at least one word including the first word while displaying the animation through the visual object.

10. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: identify a region being capable of identifying in the virtual environment corresponding to a direction viewed by a user of the electronic device, applying the animation to the visual object in the region.

11. A method performed by an electronic device comprising: displaying, via a display of the electronic device, at least part of a virtual environment including a visual object, obtaining voice information with respect to the visual object while the at least part of a virtual environment is displayed, identifying at least one word from the voice information, identifying that a first word, from among the at least one word, corresponds to a reference value, based on the first word corresponding to the reference value, changing the first word to a second word, and displaying, via the display, an animation representing uttering the at least one word including the second word through the visual object in the at least part of the virtual environment.

12. The method of claim 11, wherein the voice information includes at least one of: first information in which a user of another electronic device different from the electronic device has input with respect to the virtual object in the virtual environment, or second information displayed in a server related to a software application displaying the virtual environment, wherein the reference value includes a predetermined value by a user of the electronic device and/or a value representing an expression unavailable in the virtual environment, wherein the at least part of the virtual environment is displayed based on information received from the server, wherein the visual object represents an avatar corresponding to the user of the another electronic device.

13. The method of claim 12, comprising: receiving, from the another electronic device, a result of the first information, wherein the result includes information for representing whether the first word of the at least one word identified from the first information corresponds to the reference value representing an expression unavailable in the virtual environment.

14. The method of claim 13, comprising: identifying information representing a relationship between the user of the electronic device and the user of the another electronic device corresponding to the visual object in the virtual environment, based on the information representing the relationship being less than another reference value, identifying a change level of the animation as a first level, based on the information representing the relationship is greater than or equal to the another reference value, identifying the change level of the animation as a second level lower than the first level, displaying the animation with the identified change level through the visual object.

15. The method of claim 12, further comprising: identifying whether the first word of the at least one word identified from the first information corresponds to a predetermined value by the user of the electronic device.

16. The method of claim 11, further comprising: performing a processing of input information including the voice information, wherein the processing includes identifying the at least one word by obtaining at least one phoneme for the voice information, wherein the input information includes text information, the voice information, and video information.

17. The method of claim 11, wherein the value represented by a part of the first word includes a consonant or a vowel, wherein, in response to the part of the first word being a first consonant, portion of the second word corresponding to the part is a second consonant different from the first consonant, wherein, in response to the first word being a first vowel, portion of the second word corresponding to the part is a second vowel different from the first vowel.

18. The method of claim 11, comprising: obtaining information on a space of the virtual environment by using a software application displaying the virtual environment, identifying the animation based on the information on the space of the virtual environment.

19. The method of claim 11, further comprising outputting, via a speaker of the electronic device, the at least one word including the first word while displaying the animation through the visual

object.

20. A non-transitory computer readable storage medium, storing one or more programs including instructions which when executed by a processor of the electronic device cause the electronic device to: display, via the display, a virtual environment including a visual object, obtain voice information with respect to the visual object while the virtual environment is displayed, identify at least one word from the voice information, identify that a first word from among the at least one word corresponds to a reference value, based on the first word corresponding to the reference value, change the first word to a second word, and display, via the display, an animation representing uttering the at least one word including the second word through the visual object in the virtual environment.
